



Global Expansion of Chinese Construction Machinery: Lessons from Brazil

September 14, 2026

Executive Summary

China's exports have been strong this year, surging by 19% YoY in the first eight months. Among these exports, heavy-asset capital goods have played a particularly important role. Using construction machinery as a representative example, we illustrate how stronger product competitiveness, Chinese overseas projects, and Chinese manufacturers' "going-global" strategies have enabled market share gains across a broad range of economies.

Chinese construction machinery has expanded across the globe since 2020. Exports of related products quadrupled between 2020 and 2025 and have remained on a strong upward trend this year. More importantly, Chinese products have achieved significant market-share gains across a broad range of economies and accounted for 26% of global imports in 2025 compared to 10% in 2019. In this report, we aim to explore the factors behind this.

We use Chinese excavator exports to Brazil as a case study. Excavators are one of the most representative categories of construction machinery, while Brazil is South America's largest excavator importer and ranks among the world's top 20. Chinese excavators have also made remarkable market-share gains in Brazil, with their share rising from 20% to 90% in six years. Brazil therefore offers a compelling case study on this topic.

Through the case study and our analysis of aggregate data, we identify three drivers of the global expansion of Chinese construction machinery:

- **Product competitiveness:** Chinese brands have steadily improved both price advantage and product quality.
- **Demand from Chinese overseas projects:** Chinese contractors are undertaking more infrastructure projects and naturally exhibit stronger preferences for Chinese equipment.
- **"Going global" strategy by Chinese companies:** Higher profitability has encouraged them to direct more resources towards overseas expansion.

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Introduction

China's exports have been strong this year, surging by 19% YoY in the first eight months. In a note published for clients in March, we identified several drivers of this export growth, including AI-related demand, energy-cost advantages and the green transition. In particular, heavy-asset capital goods have been an important contributor, with construction machinery offering a representative example.

Chinese construction machinery has expanded rapidly across the globe since 2020. Using HS8429 products—including excavators, bulldozers, graders and related equipment—as a proxy, export values quadrupled between 2020 and 2025 ([Figure 1](#)). More importantly, the expansion has been broad-based: 128 of the 140 economies reporting HS8429 imports recorded an increase in China's market share over the period ([Figure 2](#)). Chinese products now account for more than half of HS8429 imports in over 30 economies. As a result, China's share of global HS8429 imports has risen from 10% in 2019 to 26% in 2025.

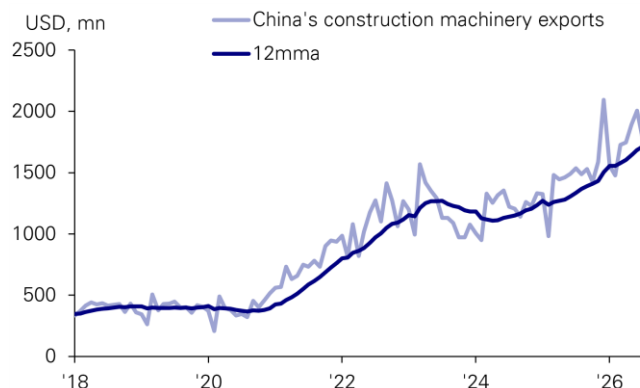
Looking at aggregate trade data alone does not fully reveal the multiple underlying drivers. At the country level, economic structures, investment cycles and infrastructure demand vary substantially, leading to very different import patterns and market-share dynamics. At the product level, HS8429 encompasses a diverse range of machinery categories with different technologies, end-markets and competitive landscapes.

Therefore, we adopt a case-study approach and focus on Chinese excavator exports to Brazil. We believe this is particularly representative since:

- Excavators are highly representative of the construction machinery industry, accounting for half of China's HS8429 exports.
- Brazil is the largest excavator importer in South America and ranks among the top twenty importers globally. The demand is underpinned by three factors. 1) As a major mining economy, Brazil requires substantial earthmoving equipment. 2) Brazil is one of the world's largest agricultural producers. The farming sector relies heavily on [capital investment and machinery](#) to raise productivity. 3) The government has [renewed its focus](#) on infrastructure development in recent years.
- Chinese excavator manufacturers have achieved exceptional market-share gains in Brazil. Their share increased from 20% to 90% over the past six years ([Figure 4](#)), with gains observed across all excavator segments.

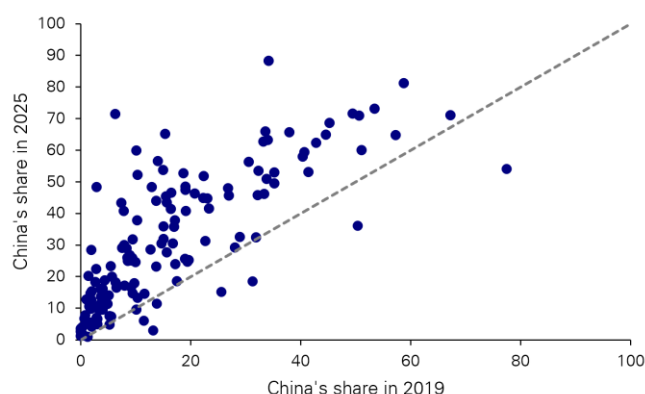


Figure 1: China's construction machinery exports have expanded rapidly since 2020



Source: Deutsche Bank Research, CEIC. We use HS8429 products—including excavators, bulldozers, graders and related equipment—as a proxy.

Figure 2: The expansion of China's construction machinery exports has been remarkably broad-based



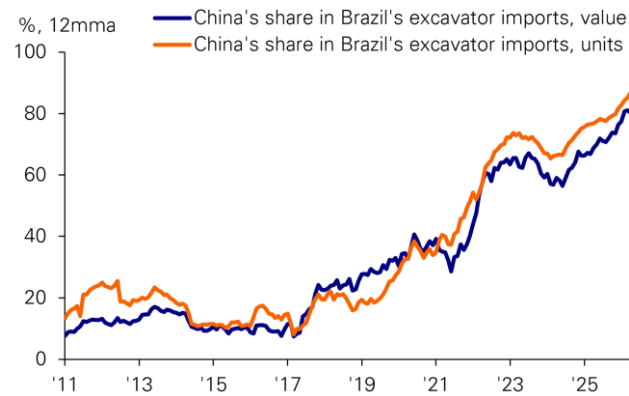
Source: Deutsche Bank Research, UN Comtrade. Each dot represents an economy.

Figure 3: Brazil has been a major excavator importer



Source: Deutsche Bank Research, UN Comtrade. Excavators are primarily classified under HS842952.

Figure 4: Chinese excavators have recorded remarkable market-share gains in Brazil



Source: Deutsche Bank Research, Brazil Custom

Expansion of Chinese Excavators in Brazil

Chinese excavators have become increasingly visible across Brazil. Their presence is no longer confined to mines, railways, and other construction sites. They are increasingly involved in landmark projects such as Rio Olympic venues and World Cup stadiums, and can even be found in residential backyards and farms. Chinese brands [reportedly](#) occupied the top three positions in Brazil's market in 2025, while six Chinese brands ranked among the top ten. Customs data paint a stronger picture: Chinese products now account for ~90% of Brazil's excavator imports, both by value and volume.

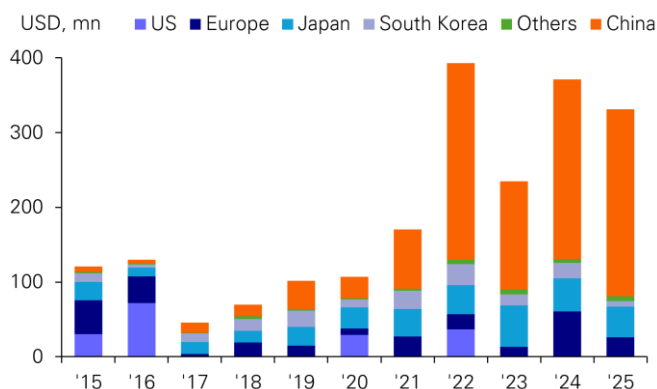
China's expansion appears to have been supported mainly by Brazil's incremental demand rather than the outright displacement of competing suppliers. During the market recovery from 2017 to 2021, excavator imports rose from all major sources. From 2022 to 2025, imports from China grew much faster, driving a rapid increase in China's market share, while annual imports from other suppliers—including Europe, Japan and Korea—remained broadly stable at around USD 100mn in total (Figure 5). This does not mean that other suppliers



were unaffected by Chinese competition: without China's sharp market-share gains, they would likely have captured a larger share of Brazil's surge in excavator demand.

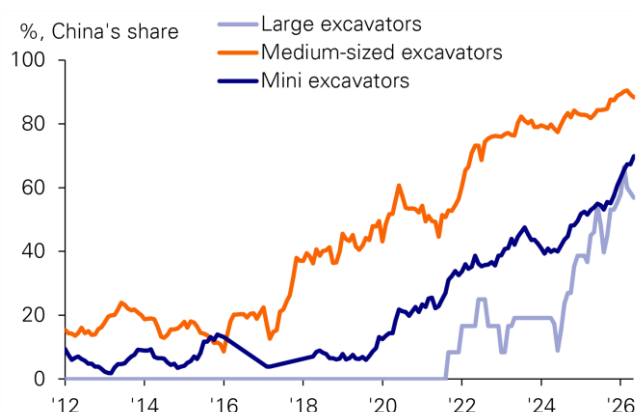
Interestingly, the expansion has been different across segments (Figure 6). Medium-sized excavators began gaining market share steadily after 2017, while mini excavators only entered a similar trajectory after 2019. Large excavators followed a different path. Prior to 2021, Chinese brands were absent, yet they rapidly gained traction after the first entry. The distinct patterns suggest that the rise of Chinese excavators might have been driven by multiple demand- and supply-side forces.

Figure 5: This expansion appears to have come largely from incremental demand rather than outright displacement



Source: Deutsche Bank Research, Brazil Custom

Figure 6: Expansion has been different across product segments



Source: Deutsche Bank Research, Brazil Custom. Brazil's 8-digit NCM customs classification distinguishes excavators by power categories, allowing us to group them into three segments.

Factor 1: Price Advantage Meets Quality Upgrading

A widening price advantage has been a major driver behind the expansion of Chinese products in the medium-sized and mini excavator segments. Since these products are relatively less technologically demanding, pricing remains a major consideration for buyers. We divide the sample into four periods and compare China's price advantage against market share gains (Figure 7). China's advantage has steadily expanded, with products now more than 50% cheaper than peers. Meanwhile, the relationship between price advantage and market share is highly significant. For medium-sized excavators, a one percentage point increase in China's price advantage is associated with a 0.4pt gain in market share. For mini excavators, the corresponding relationship is even higher at 0.9pt.

Indeed, mini excavators provide the clearest example of price-driven expansion.

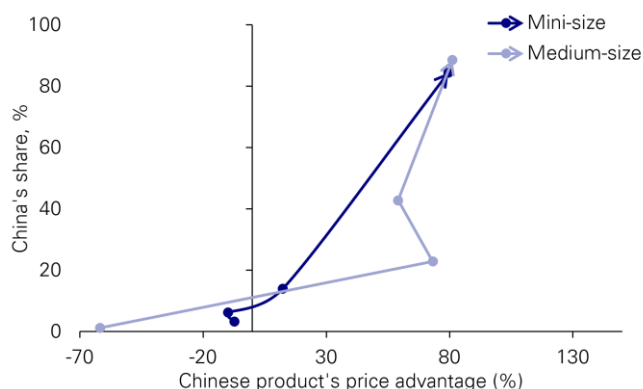
These machines are used for small-scale construction projects or purchased by individual consumers, making demand particularly price-sensitive. Chinese products began to undercut competitors after 2017, and their price advantage widened significantly after 2020 (Figure 8). The gap has now reached USD 17,000 per unit. We further decompose this price advantage (Figure 9). 60% of the widening gap reflects price increases by overseas competitors amid higher inflation, while 40% reflects price cuts by Chinese manufacturers. RMB



depreciation accounted for 10% of the gap during 2023-25. However, following its appreciation in 2026, the cumulative contribution has turned negative.

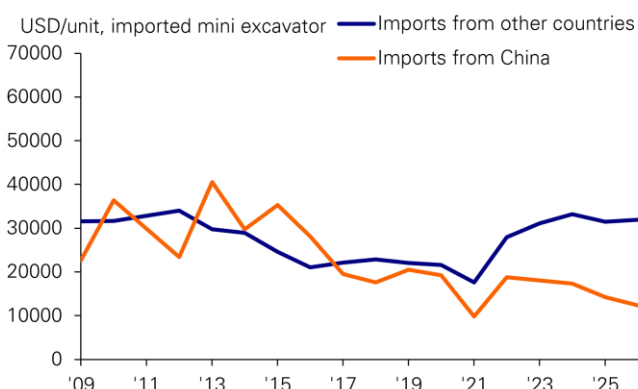
More noteworthy is the advance of Chinese brands into the high-end segment. Large excavators are far more technologically sophisticated. Before 2021, Chinese brands were virtually absent from Brazil's large excavators market. They entered the market in 2021 and expanded rapidly thereafter (Figure 10). So far in 2026, 43 of Brazil's 47 imported large excavators have come from China. Chinese manufacturers are reportedly delivering [100-tonne-class excavators](#) to Brazilian mining customers and are actively promoting equipment in the [500-tonne category](#). Chinese companies' competitiveness is increasingly extending from price advantage toward technology intensity.

Figure 7: A widening price advantage has driven expansion in the medium-sized and mini excavator



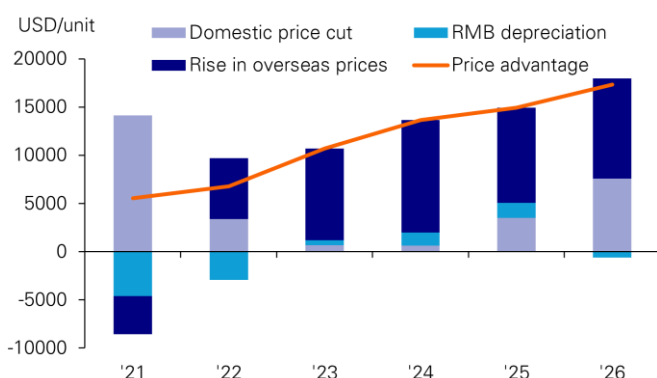
Source: Deutsche Bank Research, Brazil Custom. The four observations correspond to the periods 2006-2010, 2011-2015, 2016-2020, and 2021-2025, respectively.

Figure 8: Mini excavators provide the clearest example of price-driven expansion



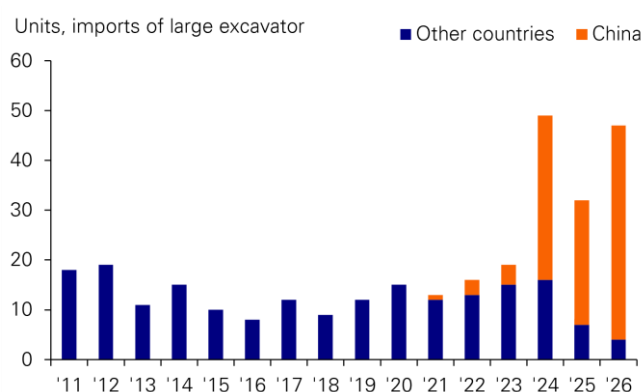
Source: Deutsche Bank Research, Brazil Custom

Figure 9: China's price advantage has widened owing to overseas inflation, domestic price cuts and RMB depreciation



Source: Deutsche Bank Research, Brazil Custom

Figure 10: Chinese brands entered the large excavator market in 2021 and rapidly expanded thereafter



Source: Deutsche Bank Research, Brazil Custom

Factor 2: Demand from Chinese Overseas Projects

Chinese firms are undertaking more construction projects in Brazil, generating additional demand. They naturally have a stronger preference for Chinese-made machinery. The value of construction projects completed by Chinese companies has risen consistently since 2021. Growth in newly signed contracts has been



even stronger, with their value reaching a record USD 4.6bn in 2024 (Figure 11). This trend may have contributed to the growth in excavator imports since 2024, especially for large excavators.

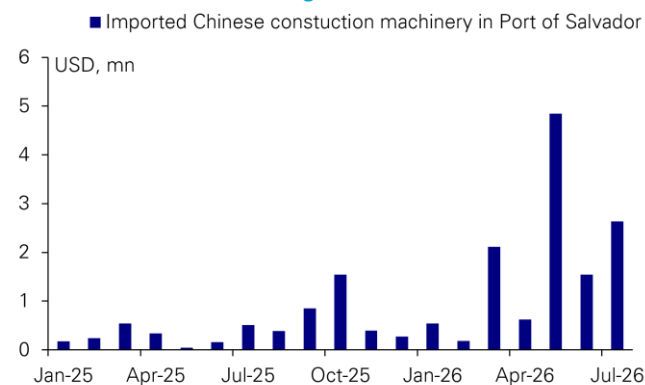
The Salvador-Itaparica Bridge project provides a compelling example. Once completed, it will be the longest sea-crossing bridge in South America. The project was awarded to two Chinese SOEs in 2019, with an initial investment of USD 2.2bn, which was raised to USD 2.7bn in 2025. It is the largest infrastructure investment undertaken by Chinese companies in Brazil. Since construction began this year, [~800 tons of equipment](#) from China have arrived at the port of Salvador since April, consistent with the surge in custom data (Figure 12). Most recently on July 1, the project company showcased the first batch of equipment deployed on site, [all sourced from China](#).

Figure 11: Chinese firms are undertaking more construction projects in Brazil



Source: Deutsche Bank Research, China Mofcom

Figure 12: Imports of Chinese machinery through the Port of Salvador have surged in recent months



Source: Deutsche Bank Research, Brazil Custom

Factor 3: Supply-Side Expansion Driven by Higher Profitability

Chinese firms have strong incentives to expand overseas because foreign markets offer higher profitability. Some views have argued that low export prices primarily reflect dumping driven by weak domestic demand. If that were the case, corporate profits would be expected to remain flat or deteriorate as firms sacrificed margins to gain market share. Company-level evidence suggests the opposite.

We examine five listed Chinese construction machinery manufacturers. Combined overseas revenue has risen consistently since 2020, doubling in 2022 and doubling again by 2025, broadly mirroring the trajectory of China's total construction machinery exports (Figure 13). More importantly, overseas profitability has improved substantially. Across the five companies, operating margins in overseas markets have increased by 5-15 percentage points, with the average rising from 20% during the previous decade to 30% today (Figure 14). In contrast, domestic margins have fallen by 5 percentage points.

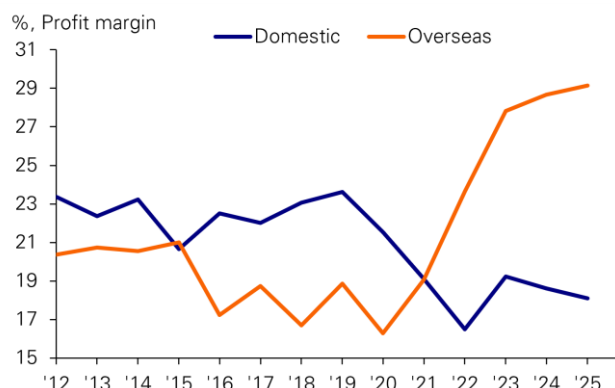


Figure 13: Overseas revenue has risen consistently since 2021



Source: Deutsche Bank Research, Wind

Figure 14: Overseas profit margins have exceeded domestic margins



Source: Deutsche Bank Research, Wind

Beyond Excavators: Chinese Construction Machinery Expansion

Through this case study, we identify three forces underpinning the expansion of Chinese excavators in Brazil: improving price and quality competitiveness, growing demand from Chinese overseas projects, and proactive going-global strategies by Chinese companies.

We believe the forces extend to the broader construction machinery sector.

- **Price and Quality Competitiveness.** The price advantage over HS8429 has become more pronounced since 2022 (Figure 15). The analysis of eight-digit categories revealed a more comprehensive picture (Figure 16). First, exports increased across all categories, including those with rising prices. Second, the products recording larger price increases tended to have higher unit values. That said, the expansion is driven by price cuts as well as increasing demand for higher-value products.
- **Overseas Expansion of Chinese Enterprises.** Chinese overseas construction projects have reaccelerated since 2024, with completed value and newly signed contracts rising by 20% and 15%, respectively (Figure 17). Belt and Road economies are key destinations for Chinese overseas construction projects, and Chinese construction machinery has gained market share much more rapidly in these markets.
- **Proactive “Going global” strategy.** Since the 2010s, construction machinery manufacturers have viewed overseas expansion as a key pillar of future growth and have invested heavily in international distribution channels, service networks and localized production facilities.

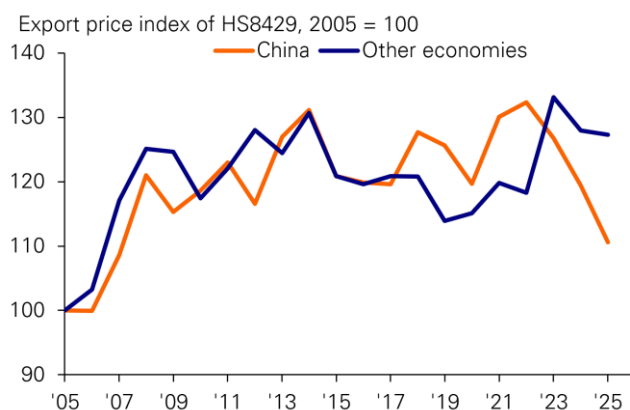
Looking ahead, we believe all three drivers are likely to remain in place for an extended period. China’s manufacturing capability and continued technological progress should support further gains in product competitiveness. Accelerating component localization could further strengthen the price advantage. Meanwhile, the overseas expansion of Chinese companies is expected to remain a long-term trend. Belt and Road economies warrant particular attention, as Chinese



investment in these markets has surged in recent years, laying a solid foundation for further overseas expansion (Figure 18). As a result, we remain constructive on the outlook for construction machinery exports.

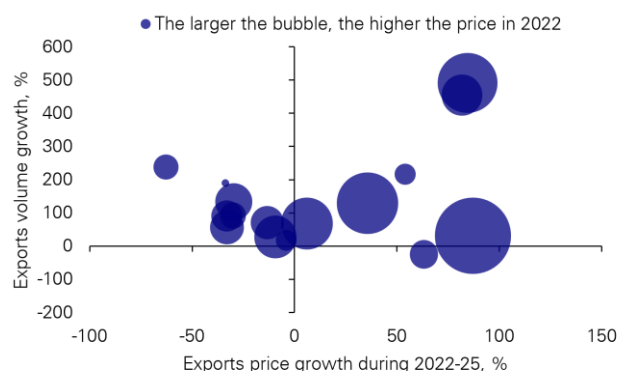
That said, several risks warrant monitoring. First, further RMB appreciation could erode the price competitiveness of Chinese products. Second, the sharp increase in market share achieved in Brazil—from 20% to 90%—may not be easily replicated elsewhere. In some markets, Chinese firms could face stronger competition from local manufacturers, limiting the pace of further market-share gains.

Figure 15: China's price advantage on HS8429 has become more pronounced since 2022



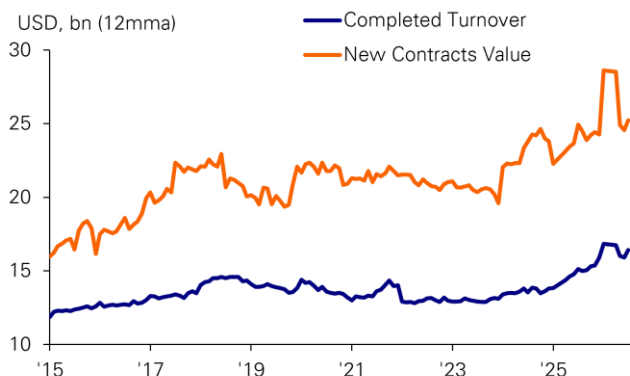
Source: Deutsche Bank Research, UN Comtrade

Figure 16: The expansion is driven by price cuts and increasing demand for higher-value products



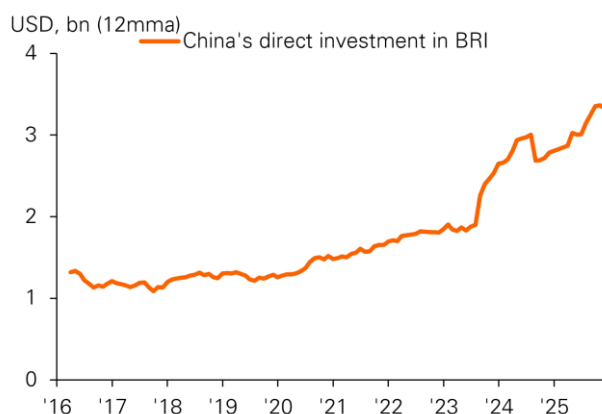
Source: Deutsche Bank Research, CEIC. Each bubble represents an eight-digit category under HS8429.

Figure 17: Chinese contractors have been undertaking more projects overseas since 2024



Source: Deutsche Bank Research, CEIC

Figure 18: China's direct investment in Belt-and-Road economies has also surged since 2024



Source: Deutsche Bank Research, CEIC



Appendix 1

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